

Final Decision made for COMMSBIO-24-7622B

commsbio@nature.com

mar. 24/03/2026 12:03

À : Jacques BOURG <jacques.bourg@pasteur.fr>;

Dear Dr. Bourg:

Here is a copy of the decision letter for manuscript "autoFISH: a modular toolbox for sequential single-molecule RNA FISH experiments" by Dr Florian Mueller and colleagues [Paper #COMMSBIO-24-7622B], which you were a Contributing Author.

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Sincerely,

Many thanks,
Kaliya Georgieva, PhD
Associate Editor
Communications Biology
<https://orcid.org/0009-0006-2251-2578>

Subject: Acceptance of COMMSBIO-24-7622B

Dear Dr Mueller,

We are delighted to accept your manuscript titled "autoFISH: a modular toolbox for sequential single-molecule RNA FISH experiments" for publication in Communications Biology. Thank you for choosing to publish your interesting work with us.

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Providing great service is very important to us. We would greatly appreciate any comments you have about your experience at Communications Biology. We look forward to publishing your paper, and we hope to work with you again in the future.

Best regards,

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autoFISH: a modular toolbox for sequential single-molecule RNA FISH experiments

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Abstract

Understanding the role of gene expression in cellular function and tissue organization requires spatial and quantitative detection of individual RNA molecules; however, the widespread adoption of automated single-molecule fluorescence in situ hybridization (smFISH) has been limited by the cost of equipment and the complexity of experimental procedures. We present autoFISH, an affordable, user-friendly platform that removes these barriers through open-source hardware components, accessible control software, and integrated analysis tools. The system demonstrates broad applicability by enabling both conventional and signal-amplified smFISH protocols and incorporates an optimized tissue-clearing method that preserves nuclear structures. Testing across multiple cell types and tissue preparations validates the system's reliability and reproducibility, offering a practical solution for scaling spatial transcriptomics research and advancing discoveries in cellular and developmental biology, while significantly reducing costs and the technical expertise required.

Introduction

Transcription is a fundamental biological process pivotal to development and disease. Individual cells exhibit considerable heterogeneity in their transcriptional profiles, including RNA abundance and intracellular RNA localization. Accurately quantifying the number of RNAs within cells and their locations provides a precise readout of a cell's molecular state and enhances our

understanding of RNA regulation. Such measurements require spatially resolved RNA detection in the native cellular environment with sub-cellular accuracy. Here, single-molecule FISH (smFISH) is widely regarded as the gold standard for detecting individual RNA molecules with high sensitivity and precise sub-micrometer localization (Femino et al., 1998). Various adaptations of this method have been developed, all based on the principle of targeting RNAs with multiple oligonucleotide fluorescent probes, either directly labeled (Raj et al., 2008) or indirectly labeled via secondary probes (Tsanov et al., 2016) or signal amplification systems (e.g. branched- DNA (Xia et al., 2019), HCR- FISH (Choi et al., 2018), SABER (Kishi et al., 2019)).

RNAs then appear as diffraction-limited spots in fluorescence microscopy images, and their positions can be measured with high spatial accuracy using dedicated analysis workflows (Bahry et al., 2021; Imbert et al., 2022; Xu et al., 2024). When combined with cell segmentation (Stringer et al., 2020), the RNA abundance in each cell can be estimated, and statistical analysis can be used to quantitatively describe subcellular RNA localization (Chouaib et al., 2020; Mah et al., 2024). Crucially, smFISH preserves the spatial context of cells while revealing their transcriptional states. This dual output facilitates advanced quantitative analysis of tissue architecture by integrating molecular profiles, cellular identities, and spatial information. Cell types can be identified using standard scRNA-seq clustering approaches (Traag et al., 2019), and their annotation can be further refined by incorporating spatial data (Teng et al., 2022). This enables large-scale investigation of features such as cell-cell communication, tissue composition, and complex tissue gradients (Palla et al., 2022).

A significant limitation of the original smFISH methods is that the number of spectrally distinguishable fluorophores constrains the number of distinct RNA species that can be visualized simultaneously. To overcome this limitation, new methods have been developed that rely on iterative hybridizations, i.e., alternating cycles of probe hybridization, imaging, and signal removal (Chen et al., 2015; Codeluppi et al., 2018; Lubeck et al., 2014). A general feature of these approaches is that the primary probes carry one or more readout sequences that can be detected by a secondary, complementary readout oligo. Usually, primary probes against all RNA targets are hybridized on the bench, while the readout oligos are hybridized directly on the microscope with an automated fluidic system. Signal removal can be achieved by various methods; the most commonly used are stripping with high-stringency buffers (Lohoff et al., 2022), fluorophore removal via chemical cleavage (Moffitt et al., 2016b), or displacement with a tertiary stripping oligo (Mateo et al., 2021). In sequential smFISH experiments, the number of targeted species is linearly correlated with the number of iterations and the number of fluorophores used. Multiplexed barcoding approaches have driven the field of spatial transcriptomics by dramatically increasing RNA detection throughput (Cheng et al., 2023). Here, each RNA species carries several readout sequences forming unique RNA-specific barcodes. Although readout sequences are reused across RNAs, each hybridization round visualizes a different set of genes. RNA identities must be decoded post-analysis. When combined with error correction, this combinatorial method enables the detection of hundreds to thousands of RNA species.

While several commercial systems offer convenient turnkey solutions with increasingly large panel sizes, they are costly in terms of instrumentation and reagents and lack flexibility.

Exploratory analyses, time courses, dose-response studies, and replicates thus demand substantial financial resources. While large RNA panels are often necessary, not all studies require such depth; smaller, more flexible panels may suffice. For example, a small probe panel with 10 genes can be adequate to resolve the main cell types in tumor cryosections (Rademacher et al., 2024).

While these commercial methods are user-friendly, they often lack flexibility across microscopy (e.g., imaging modalities) and experimental protocols (e.g., RNA detection approaches). This limitation can be critical for challenging samples, such as tissue sections, where weak RNA signals benefit from amplification to improve detection reliability. Several widely used amplification strategies include branched DNA (Wan et al., 2024; Xia et al., 2019), hybridization chain reaction (HCR) (Choi et al., 2018; Gandin et al., 2025), and SABER (Kishi et al., 2019). SABER is appealing due to its simple base protocol and tunable amplification levels.

Taken together, home-built solutions remain a crucial option for addressing many research questions. To address this need, we propose autoFISH, an end-to-end open-source workflow comprising hardware, software, and experimental protocols for automated smFISH experiments designed for flexibility and affordability. The autoFISH toolbox relies on a versatile and affordable fluidic system, for which we provide detailed building plans. Flexibility comes in several flavors. First, the imaging modality can be chosen depending on the desired application, as the fluidics system required for sequential hybridizations can be mounted on various microscopes. Second, protocols can be readily modified to meet experimental needs, such as by incorporating signal amplification or tissue clearing. Lastly, smaller-scale experiments can be readily performed because probe design and synthesis are amenable to control.

We demonstrate the applicability of autoFISH on two experimental datasets. Firstly, HeLa cells, which we use to benchmark the performance of the protocol and illustrate its flexibility by alternating between standard and amplified FISH experiments with SABER (Kishi et al., 2019), and secondly, mouse lung tissue, which we use to visualize several marker genes from our earlier study (Curras-Alonso et al., 2023). Here, we introduce a modified tissue-clearing protocol that improves nuclear signal. In summary, we provide a comprehensive toolbox for establishing automated, sequential smFISH experiments at a lower cost.

Results

Here, we present autoFISH (Figure 1), a universal toolbox for performing sequential smFISH, with instrumentation that can be built in-house at an affordable cost and adapted to specific needs and experimental protocols using standard buffers. autoFISH is based on several previous studies (Bintu et al., 2018; Imbert et al., 2022; Kishi et al., 2019; Moffitt and Zhuang, 2016; Tsanov et al., 2016) and is entirely open-source. We also provide Python workflows to analyze these data and infer single-cell RNA quantification. autoFISH is hosted on GitHub (see Methods) and includes detailed build plans, control software, and experimental protocols. To validate and demonstrate the capability of autoFISH, we implemented and validated different protocols, including workflows for signal amplification and tissue clearing on both cultured cells and tissue sections.

Computer-controlled fluidics system for automated experiments

A sequential FISH experiment consists of several key steps: hybridization of primary oligos overnight at 37 °C, sample mounting on the microscope, and alternating imaging and incubation steps for fluorescently labeled secondary probe hybridization and displacement of secondary probes from the previous round (Figure 1A).

The core of autoFISH is a computer-controlled fluidics system that performs the sequential hybridization steps and automates imaging (Figure 1B). This system enables flushing buffers from syringes, providing either large buffer reservoirs or single-use buffers stored in multiwell plates, which can be accessed via a pipetting robot. A computer-controlled valve allows buffer selection, and flow control is achieved via a peristaltic pump, with flow rates monitored by a flow sensor. Lastly, air bubbles are removed from the buffers with a bubble trap.

Samples can be mounted in an imaging chamber, which is ideal for single-condition cell culture or tissue samples (Figure 1C). We emphasize that placing the pump immediately after the valve yielded better performance than placing it at the end of the fluidic system (after the imaging chamber). Based on our experience, this resulted in higher bubble-trap efficiency and reduced deformation of the imaging chamber during pump action, which can otherwise lead to loss of microscope autofocus.

We implemented a modular, open-source Python package that controls all components of the fluidic setup to perform automated buffer exchanges (Figure 1D). The code's modularity facilitates adding new components from other providers, such as a different peristaltic pump. The user can flexibly define fluidics runs in a human-readable text file that contains all buffers, pump durations, and incubation times, as well as optional steps, such as performing DAPI staining only for one run or adding hybridization steps, e.g., for signal amplification with SABER.

For fully automated experiments, imaging must be automated as well. As a design choice, we do not directly control microscopes within our code; instead, we provide methods to interface with acquisition software packages. Currently, we support acquisition in Micromanager via Pycromanager, or triggered acquisition via a TTL pulse or a shared synchronization text file (see Methods for details). With these solutions, autoFISH can already be used with various imaging systems, and new trigger approaches can be added as needed.

We provide a graphical user interface (GUI) to facilitate usage. This interface can be used to control the entire workflow: load the configuration files, initiate the fluidics system, trigger the microscope, and then launch either an individual fluidics run or a fully automated run, in which imaging and fluidic runs are performed sequentially and iteratively.

autoFISH with 20 hybridization rounds

We first validated the robustness of autoFISH by assessing its ability to detect RNA and remove signals. In sequential smFISH, RNA detectability is determined by several key performance

metrics. Probe stripping efficiency is fundamental and is evaluated by comparing images of the same target RNA before and after stripping. The absence of residual signal in post-stripping rounds is critical to preventing false positives in subsequent hybridization cycles. Furthermore, consistency in spot counts across identical rounds serves as a crucial metric; a stable number of detected RNA spots indicates reproducible detection efficiency throughout the detection rounds. This is complemented by cross-round colocalization analysis, which directly assesses whether the same individual transcripts are reliably identified across cycles. A high colocalization rate (e.g., ~90%) confirms the consistent detection of the same RNA target. To ensure the technical quality of the images, we monitored the signal-to-noise ratio (SNR), defined as the spot intensity relative to mean local background fluctuations. While a stable SNR is essential for reliable automated detection, tracking absolute signal and background intensities provides more detailed insight. For instance, a proportional decrease in both RNA signal and background intensity—resulting in a stable SNR—suggests a minor reduction in probe binding without compromising overall detectability. Collectively, these metrics allow evaluation of whether the system maintains robust detection across extensive multiplexing sequences.

To validate our approach, we performed a 20-round experiment in which we alternated the detection of two genes with distinct subcellular localizations (Chouaib et al., 2020): *XPO1* (randomly distributed) and *KIF1C* (localized to cell protrusions). We consistently detected the same RNA species across successive rounds (Figure 2A). Furthermore, the absence of residual signal following stripping confirms the efficiency of probe removal (Figure 2B). Quantitative analysis revealed a stable number of detected spots per round (Figure 2C) with high spatial conservation; approximately 90% of spots colocalized across rounds (Figure 2D-E and Supplementary Figure 1), matching benchmarks for dual-color RNA labeling (Tsanov et al., 2016). While the SNR remained constant throughout the experiment (Figure 2F), closer inspection indicated that successive washing cycles slightly reduced both the RNA spot intensity and the background (Figure 2G-H). Collectively, these results demonstrate robust, reproducible RNA detection over 20 hybridization rounds.

Flexible signal amplification

In standard smFISH, each RNA molecule is targeted by a pool of primary oligonucleotide probes, resulting in a locally concentrated fluorescent signal at the transcript location. To enhance detection efficiency—particularly in samples with high autofluorescence or low transcript abundance—autoFISH employs a multi-layer signal-amplification strategy.

In the basic **autoFISH** workflow, signal amplification is achieved through a modular, three-component assembly based on the framework described by (Bintu et al., 2018). Primary probes are synthesized with distinct readout sequences at both the 3' and 5' ends, which serve as binding sites for intermediate readout oligonucleotides during detection. To maximize signal while maintaining cost-efficiency, imaging oligonucleotides—each conjugated with two fluorophores—are pre-hybridized to these readout probes. This configuration enables the recruitment of up to **four fluorophores per primary probe**, enhancing the signal-to-noise ratio (SNR). While

brightness can be further increased by directly labeling the intermediate readout probes, the standard configuration typically provides sufficient sensitivity while minimizing reagent costs.

SABER offers a robust, programmable alternative (Kishi et al., 2019) by in vitro extension of oligonucleotides with repeated sequences before use in hybridization (Figure 3A). These extended oligos can then be bound by multiple imaging probes, with signal intensity precisely controlled by the amplification reaction duration. A key advantage of this pre-hybridization amplification is that it does not extend the actual smFISH experiment. To simplify this approach, we also evaluated synthetic oligos containing three repeat binding sites—the practical limit for ~150 nt synthesis. We further enhanced signal strength by labeling imaging oligos at both the 3' and 5' ends. To simplify this approach, we evaluated synthetic oligos containing three repeat binding sites—the practical limit for ~150 nt synthesis. We further enhanced signal strength by labeling imaging oligos at both the 3' and 5' ends. While signal intensity scaled with the number of binding sites, dual-end labeling yielded less than a twofold increase, likely due to steric hindrance or quenching (Supplementary Figure 1). These results suggest that for moderate amplification, synthetic oligos with three repeats provide sufficient signal enhancement.

We integrated SABER into autoFISH by replacing the imager sequence on the readouts with the SABER PER sequence required for amplification (Figure 3A). We then optimized the protocol to be compatible with the fluidics system. In short, we replaced the formamide from the original buffers with the less toxic Ethylene Carbonate. We also hybridized the SABER oligos in two steps on the fluidics system: first, the amplicons, then the imaging oligos. We performed the hybridization of the amplicons in a more stringent buffer, resulting in a substantial decrease in background. We also found that adding even a moderate amount of dextran (5% was compatible with our fluidics system) reduced the amount of imaging oligo required, thereby reducing experimental costs. With these changes, SABER can be performed on the fluidics system. Interestingly, the flexibility in the fluidics control allows us to perform SABER hybridization, which requires an additional hybridization step, in selected runs only, as demonstrated in the validation experiments.

We then performed an experiment targeting two genes (*XPO1* and *NFKBIA*), alternating between normal and SABER hybridization, and included a control round in which we only stripped SABER without a new hybridization (Figure 3B). Qualitative inspection of the images shows clear amplification of SABER for both genes, with the same RNAs visible and SABER resulting in increased signal intensity (Figure 3C). Importantly, the SABER amplicon can also be efficiently stripped and the signal removed (Figure 3C). A more detailed analysis confirmed this qualitative impression, showing a constant amount of detected RNA per round (Figure 3D). Analysis revealed that SABER substantially increased RNA signal intensity, albeit with a marginal increase in background levels (Figure 3E). Importantly, co-localization between detection rounds (both standard and SABER) for the same gene is high (Figure 3F). In contrast, co-localization between the two genes is very low, indicating no cross-detection.

These experiments demonstrate the integration of SABER into the autoFISH workflow to amplify RNA signal intensity. Furthermore, they highlight the control software's flexibility, which allows

adjustment of hybridization conditions across specific imaging runs. This combined approach enables high-sensitivity detection while maintaining the automated throughput required for complex multiplexing cycles.

autoFISH on cleared mouse lung tissue

In a recent study, we performed smFISH of selected RNA marker genes on mouse lung tissue (Curras-Alonso et al., 2023). While imaging quality was sufficient for analysis, we also encountered substantial background fluorescence. It has been demonstrated that this background arises from both tissue autofluorescence and nonspecific probe binding to various cellular structures (Moffitt et al., 2016a). Tissue-clearing methods help to efficiently reduce this background. Here, the biological sample is embedded into a nonswellable polyacrylamide (PA) matrix (Figure 4A). Several strategies exist to anchor RNAs to the PA, either through chemical compounds (Chen et al., 2016) or through the more widely used approach of oligos targeting the poly(A) tails of mRNAs that carry a terminal acrydite moiety (Moffitt et al., 2016a). The tissue can then be treated with Proteinase K and SDS to remove proteins and lipids, which are significant sources of autofluorescence and nonspecific binding (Moffitt et al., 2016a).

When applying the protocol with the Poly-T anchoring oligos to lung tissue, the background was reduced, and RNAs were detected. However, we found that the nuclear signal was often less distinct than in samples anchored with epoxide (Figure 4B). For samples such as lung tissue with a locally high nuclear density, this could adversely affect nuclear segmentation. We hypothesized that this could be due to reduced chromatin constraints after digestion. In a recent study, an epoxide (GMA) was introduced as a universal linking agent in Expansion Microscopy (ExM) for proteins and RNA (Cui et al., 2023). GMA possesses two reactive sites, allowing covalent linkage of various functional groups—particularly amine moieties—to the acrylamide gel matrix. Since the polymers used for clearing and ExM are similar, we speculated that GMA could also be used in clearing protocols and serve as a structural scaffold.

We compared DAPI staining of HeLa cells after clearing under three conditions: no anchoring, anchoring with GMA, and anchoring with GMA plus poly-T oligos. When GMA was used—either alone or with poly-T anchor oligos—nuclear staining appeared crisper (Figure 4B). To provide a more comprehensive comparison, we analyzed these images using a quantitative sharpness metric (see Methods, (Helmlí and Scherer, 2001)). This analysis confirmed our qualitative observations and demonstrated increased sharpness of the DAPI signal upon addition of GMA (Supplementary Figure 2).

We then performed autoFISH in cleared mouse lung tissue against 6 targets from our previous study (Curras-Alonso et al., 2023). Individual RNA molecules for each target were successfully detected (Figure 4C). Because dedicated membrane markers were not used, we applied our analysis package, ComSeg (Defard et al., 2024), to segment cells using nuclear staining and to detect RNA point clouds (Figure 4D). The resulting gene expression profiles then enable clustering (Figure 4E) and mapping the identified cell types back onto the image, revealing their spatial locations.

In summary, we established an optimized tissue-clearing protocol incorporating GMA to improve retention of nuclear signals. We validated this method in mouse tissue, demonstrating robust detection across a panel of marker genes.

Discussion

smFISH is a powerful technique for visualizing individual RNAs within their native cellular context. By performing multiple cycles of probe hybridization, imaging, and probe removal, the number of RNA species that can be analyzed is greatly increased. However, this also requires more complex protocols and specialized equipment. Although commercial solutions are available, they are costly and lack flexibility. Here, we introduce autoFISH, a modular toolkit for automated, sequential FISH that leverages validated, optimized workflows.

In this work, we demonstrate the performance of autoFISH across cell lines and tissues, including up to 20 hybridization runs. While the throughput of this method is lower than that of commercial multiplexed solutions, it offers some advantages. First, because each RNA is imaged individually, spatial crowding is not a concern, thereby allowing even highly expressed genes to be targeted. Additionally, by ordering oligos for each gene individually, we can customize the selection of targeted genes. One of the main strengths of autoFISH is its flexibility. We demonstrate this by applying various experimental protocols, showcasing flexible control over fluidic runs via optional steps, and by implementing different methods to trigger acquisitions on the microscope.

For more challenging samples, we provide experimental protocols for signal amplification and tissue clearing that are compatible with the autoFISH fluidic system. We illustrate how signal amplification can be achieved with SABER and demonstrate that it can be performed on specific runs via the flexibility of the fluidic control. The proposed use of epoxide in tissue clearing enhances preservation of nuclear signals, promising to improve segmentation in densely packed samples. Additionally, the epoxide used (GMA) is inexpensive, allowing for cost-effective experiments. Finally, epoxide anchoring is more versatile than targeting poly(A) tails, which limits its application to mature mRNAs. This versatility suggests that other RNA species, such as nascent or long non-coding RNAs, could potentially be anchored using this method, thereby reducing the likelihood of their loss during the clearing step.

The autoFISH control software features an open, modular architecture designed for integration with diverse imaging systems. Implemented in Python, the software provides a flexible framework for integrating various acquisition triggers and hardware components, such as valves and pumps. This open-source approach facilitates the future integration of Python-based libraries for real-time quality control and data processing.

In summary, autoFISH significantly lowers the barrier to entry for research groups implementing sequential smFISH. While commercial platforms excel at large-scale, high-multiplexing discovery experiments, in-house solutions such as the one presented here provide a cost-effective complement. By utilizing high-throughput data—such as scRNA-seq or large-scale spatial

screens—to identify refined gene signatures, researchers can use autoFISH to probe specific targets at a substantially reduced cost. Finally, as an open-source platform, autoFISH is designed for community-driven expansion, such as the development of new modules and the integration of diverse experimental protocols.

Methods

This section provides an overview of the methods. For detailed, up-to-date information, readers are referred to the following resources: <https://github.com/fish-quant/autofish> (for the control code system), <https://doi.org/10.5281/zenodo.17965896> (for fluidics building plans and experimental protocols), and <https://github.com/fish-quant/autofish-analysis> (analysis code).

Instrumentation and control

AutoFISH relies on an automated fluidics system for buffer exchanges. Python control software was developed to operate the fluidics system and communicate with the microscope to initiate acquisitions. AutoFISH is implemented as a lightweight Python package with minimal dependencies. It utilizes PySerial for communication with various hardware components, and its modular design enables the easy addition of new components, such as pumps or valves from different suppliers. AutoFISH can be controlled with a graphical user interface.

Microscopy

AutoFISH does not allow direct control of microscopes but provides several options for triggering image acquisition. This makes autoFISH highly flexible, as it can be used across various microscopes with minimal adaptation. At the time of publication, the following options are available, but new ones can be easily added:

Pycromanage (Pinkard et al., 2021) is suitable for microscopes controlled by Micromanage. The user can define a settings file with the relevant acquisition parameters and a position list. autoFISH will then create a multi-D acquisition event that enables imaging after each hybridization round.

Synchronization with a text file: either the file's content (e.g., a switch from 0 to 1) or its existence can be used to start an acquisition. Once the acquisition is complete, the file content can be set back to 0, or the file can be deleted, signaling to autoFISH that the next hybridization round can begin.

Communication with TTL trigger: This is suitable for microscopes that integrate incoming and outgoing TTL trigger events into their acquisition routines. In autoFISH, we use an Arduino to send and receive TTL triggers: a TTL trigger initiates an acquisition, and an incoming TTL trigger signals that the acquisition is terminated and that the next hybridization rounds can begin.

Fluidics system

We use a custom-built fluidics system that combines elements from published approaches (merFISH and ORCA) (Bintu et al., 2018; Moffitt and Zhuang, 2016). Detailed building plans and part lists are on GitHub (<https://doi.org/10.5281/zenodo.17965896>). The system consists of the following main components, which are widely available and affordable: a peristaltic pump to accurately control flow rate, a bubble trap connected to a vacuum pump to remove air bubbles, and a flow sensor to monitor flow and detect potential issues, such as obstructed fluidic lines or large air bubbles.

Buffers are stored either in deep-well 96-well plates (each well holding up to 2 mL) or in syringes. The former usually contains hybridization buffers used only once, and the syringes contain general-purpose buffers for washing and imaging. The 96-well plates are placed on a pipette robot built from a CNC milling machine. These robots can be readily adapted by incorporating a 3D-printed insert into the spindle holder to accommodate a needle connected to a fluidic line. The needle can aspirate liquid from the multi-well plates. A computer-controlled valve selects buffers from different inlets from the fluidic lines connected to the pipette robot or the syringes.

Sequential smFISH: instrumentation and microscopy

Detailed experimental protocols are available at <https://doi.org/10.5281/zenodo.17965896>.

In experiments on the fluidic system, we used the less-toxic Ethylene Carbonate (EC) instead of formamide, as previously described (Moffitt et al., 2016a). In most previous studies, hybridizations on the fluidics system were done without Dextran. For some experiments, such as signal amplification with SABER, we found that even low concentrations of dextran help reduce oligo concentrations, improve signal, and decrease background. We found that, with our peristaltic pump (Ismatec Reglo Digital), buffers containing up to 5% dextran can be used without detrimental effects on flow rates. It is essential to filter all buffers (using 0.22 μ m filters) to prevent clogging of the fluidic lines.

Two different microscopy systems were used in this study. Data for Figures 1, 2, and 4 were acquired on a Nikon Ti Eclipse equipped with a Lumencor Spectra X LED light source, a 60 $\times$ 1.4 NA objective, and a Prime BSI Express sCMOS camera, and dedicated monoband filters from Semrock (DAPI-5060C-OMF, Cy3-4040D-000, Cy5-4040D-000). The system was controlled via Micro-Manager, with automated acquisition through Pycro-Manager in autoFISH. Data for Figure 3, and Supplementary Figures 1 and 2 were acquired on a Leica Thunder (Dmi8) with a Leica K8 camera, 613X 1.4 NA objective, Leica LED8 light source, Leica quad-band DFT51011 filter set with a motorized emission wheel containing 460/80, 535/70, 590/50, 642/80, and 100% open positions. The system was operated using Leica LAS X software with the trigger module, and automated acquisition was achieved via TTL signals exchanged between an Arduino and the microscope through the autoFISH program.

Sequential smFISH: Probe Design and Signal Amplification

In this study, we employed the labeling strategy described by Bintu et al. (2018), which uses four oligos (Figure 1): primary oligos, readout oligos, imager oligos, and displacement oligos. For each gene, 20-50 primary oligos are designed, consisting of two parts. The first part is complementary to a portion of the target gene's RNA. The other part is an identical readout sequence placed on 3' and 5' of each oligo (provided in (Bintu et al., 2018)). All oligos targeting the same RNA share this barcode. Note that these primary oligos are not fluorescently labeled. A readout oligo with the complementary barcode sequence is then used to detect a specific RNA. This oligo is pre-hybridized with an imager oligo carrying two fluorophores. Each primary oligo is thus labeled with up to 4 fluorophores. This design is cost-effective because the same imager oligo can be used with different readout oligos. A third region on the readout oligo—the toehold—can be used to remove the fluorescent signal. Using a displacement oligo complementary to the toehold and readout sequence, the readout oligo can be efficiently stripped off, and the resulting oligo duplex can be washed out. In the experiment, all primary oligos are hybridized simultaneously during a primary incubation step, while readout oligos are hybridized sequentially to image each RNA species. Multiple RNA species can be imaged simultaneously using imager oligos conjugated to spectrally distinct fluorophores. Interestingly, displacement oligos to remove readout oligos from previous rounds can be hybridized simultaneously with new readout oligos. This allows for faster experiments.

We observed that some of the original readout sequences yielded lower signal intensities in RNA smFISH. We developed a simple strategy to systematically test their quality (Supplementary Figure 3). In short, we used a probe set against a housekeeping gene carrying a validated readout sequence (RO1). We then design an intermediate oligo with a 5' region complementary to RO1 and a 3' region complementary to the readout to be tested. The to-be-tested readout can then be hybridized with an imager oligo and visualized. This approach eliminates the need to synthesize a reference probe-set for each new readout, but requires only a simple adaptor oligo. For an up-to-date list of validated readout and imager sequences, please refer to <https://doi.org/10.5281/zenodo.17965896>.

SABER provides a simple, programmable method for signal amplification (Kishi et al., 2019). We successfully achieve up to a 10-fold amplification without inducing large oligo aggregations. Specific hybridization rounds can be amplified and stripped by using the amplified SABER sequences instead of the imager sequences on the readout oligos. During SABER optimization, we observed abundant nonspecific dots arising from either nonspecific primary probes or amplicons, likely due to signal amplification. To reduce this background, we found that a 3-step hybridization protocol performs best when stringency is optimized at each step. First, primary probes are hybridized on the bench, using a more stringent buffer (1X SSC, 30% formamide, 10% dextran) rather than our standard buffer (2X SSC, 20% formamide). Second, the amplicons are hybridized in the fluidics system for each round using 1xSSC, 20% EC, and 5% dextran. Lastly, the imaging oligos are hybridized on the fluidics system (2X SSC, 10% EC, 5% dextran). The melting temperature of these oligos is relatively low because they contain no Guanine, and a more

stringent buffer results in a loss of signal intensity. The low dextran concentration does not affect flow rates, allowing a reduction in the concentration of the imaging oligos.

Target-specific oligonucleotides can be designed using various computational frameworks (Defard et al., 2025). For this study, we utilized Oligostan (Tsanov et al., 2016), a tool we developed specifically for the bioinformatic design of smFISH probes. All probe sequences are available in Supplementary File 1.

The choice of synthesis method is typically dictated by the required probe scale. For small-scale applications involving several hundred oligonucleotides, we utilize pre-synthesized oligo pools (e.g., IDT oPools) at a concentration of 50 pmol per oligo. These pools do not require further amplification and provide sufficient material for several hundred hybridization rounds on 12 mm coverslips. This approach enables modular assembly of custom hybridization panels tailored to specific biological questions. For larger-scale experiments requiring higher probe counts, we employ complex oligo pools that necessitate enzymatic amplification prior to use (Moffitt and Zhuang, 2016).

Clearing

Sample clearing reduces background signal in smFISH by removing tissue autofluorescence and reducing probe off-target binding to lipids and proteins (Moffitt et al., 2016a). Briefly, RNAs are first anchored to a polyacrylamide matrix using poly (A)-targeting oligos and/or an epoxide (GMA). Once the matrix has solidified, tissues are incubated overnight at 37°C in a Proteinase K solution (20mg/mL). The next day, tissues are washed, and hybridization of readout oligos is performed.

Sequential smFISH in HeLa cells

To validate the autoFISH workflow, HeLa cells (ATCC CCL-2) were cultured in DMEM (Gibco 21063-029), supplemented with 10% FBS and 1% penicillin-streptomycin at 37°C with 5% CO₂. Cells were then fixed in 4% PFA and permeabilized with 70% ethanol at -20°C overnight. After rehydration, the samples were hybridized overnight with primary probes against *XPO1*, *KIF1C* (Figure 2), or *XPO1* and *NFKBIA* (Figure 3). Following mounting on the appropriate support, the samples were installed on the fluidics system. For the 20-round experiments, a series of 20 alternating hybridizations and stripping cycles for *XPO1* and *KIF1C* was then performed on the fluidic system. This enabled monitoring of signal quality over time, observing probe dehybridization, and tracking its progression throughout the experiment. For the SABER experiments, the experiments were performed as described above.

Mouse lung tissue

Studies were performed in accordance with the European Community's (2010/63/UE) recommendations for the care and use of laboratory animals. Experimental procedures were specifically approved by the ethics committee of the Institut Curie, CEEA-IC #118 (Authorization

number APAFIS#5479-201605271 0291841, given by the National Authority), in compliance with international guidelines. Ten to sixteen-week-old C57Bl6/J female mice were housed in the Institut Curie animal facility in compliance with European ethical procedures. Before tissue processing, mice were euthanized by cervical dislocation. To validate the hybridization of several genes on a section of fixed and cleared tissue, autoFISH was performed on mouse lung tissue. Briefly, mouse lungs were inflated with cold 4% paraformaldehyde (PFA), fixed overnight at 4°C, cryopreserved in 30% sucrose, and embedded in optimal cutting temperature (OCT) compound before sectioning (Curras-Alonso et al., 2023).

Analysis: Sharpness measurements

3D DAPI image stacks were acquired and projected along the z-axis using maximum intensity projection. Nuclei were segmented from the projected images with CellPose (Stringer et al., 2020). For each nucleus, we determined its bounding box and computed the local contrast of every z-plane in the 3D images using Helmlí's and Scherer local contrast algorithm (Helmlí and Scherer, 2001; Tsanov et al., 2016) and Supplementary Note 1. For each nucleus, the plane with the highest focal contrast was selected for subsequent analysis. Composite images were then created by including the optimal focal-plane image for each nuclear mask, and background pixels were filled with the median z-index across all cells. Finally, we constructed box plots summarizing the extracted focus measurements for all nuclei in each condition.

Analysis: Sequential smFISH

Raw images are first reformatted into individual Z-stack .tif files, each corresponding to a field of view and hybridization round. To account for mechanical drift that may occur over successive cycles, we perform **spatial registration** using the **SimpleITK** package (Beare et al., 2018). This process uses image-similarity metrics to align images across rounds, ensuring that RNA spots and cellular structures remain spatially aligned throughout the experiment.

For analysis of cell culture experiments, spots are detected using Big-FISH (Imbert et al., 2022). Big-FISH enhances punctate signals within the images using a Laplacian-of-Gaussian (LoG) filter, followed by the identification of RNA spots as local maxima. For lung tissue experiments, spot detection was performed using the recent deep-learning approach U-FISH (Xu et al., 2024). U-FISH replaces Big-FISH's LoG filter with a deep-learning U-Net signal amplifier. It enhances the true RNA spots signal and suppresses artefact signals. This U-Net feature is particularly advantageous for complex tissues such as the lung. Spot quality metrics, such as signal-to-noise ratio, background, and spot intensity, were computed using Big-FISH (Imbert et al., 2022). Nuclei segmentation on DAPI and cell segmentation, when cell staining was present, are performed with Cellpose (Pachitariu and Stringer, 2022). Finally, all fields of view are stitched using the stitching ImageJ plugin (Preibisch et al., 2009). The final step is to generate the spatially resolved gene-by-cell expression matrix from the previously localized RNA spots. However, because cell boundary staining is not always available in our experiments, the cell expression matrix is computed using our recently published point cloud segmentation method, ComSeg (Defard et al.,

2024). To perform cell type calling, we calculated the cosine distance between the cell expression vectors and the median centroids of each cell type, as defined in the reference scRNA-seq data from (Curras-Alonso et al., 2023). These gene markers allow for the classification of five distinct cell types: Alveolar Macrophage (AM) with *Chil3*, Alveolar Type 1 (AT1) with *Rtn2*, Alveolar Type 2 (AT2) with *Lamp3*, and Epithelial Cells with *Pecam1* and *Ptpnb*. Cells are assigned to their nearest cell type cluster based on cosine distance. Given the limitations of our gene markers, not every cell can be classified. Therefore, we only classify cells with a cosine distance to the nearest cell-type centroid below 0.8 and with more than 10 detected RNAs. The analysis code is available at <https://github.com/fish-quant/autofish-analysis>.

Analysis: Image registration and RNA colocalization for SABER

To align image sets across rounds, in the absence of a dedicated fiducial marker, we generated binary spot maps and used phase cross-correlation to compute and apply optimal global shifts (ΔX , ΔY). This shift is then applied to the spot coordinates of rounds (*XPO1*; 3 and 7) and (*NFKB1A*; 5 and 8) to precisely align them with their respective reference rounds 1 and 2. We then performed a secondary manual alignment using a Napari plugin (Gros et al., 2025) between *NFKB1A* (round 2) and the primary reference *XPO1* (round 1). The resulting translation is then applied to all rounds of *NFKB1A* (2, 5, and 8). Finally, spots were matched across rounds using the Hungarian algorithm to minimize inter-round distances. Colocalization was defined as a separation of <2 pixels ($\approx 2\mu\text{m}$) and reported as the percentage of total spots identified in that round.

Statistics and Reproducibility

Consistent with proof-of-concept technology-development studies, validation relied on extensive technical sampling rather than biological replication. The statistical unit (n) corresponds to individual RNA molecules/spots, nested within cells and fields of view (FOVs). For each detected RNA, quantitative metrics including spot intensity, signal-to-noise ratio (SNR), local background, and co-localization scores (when applicable) were computed. Specific n values for cells, RNA counts, and per-molecule measurements are provided in the figure legends. Data are reported as Median $\pm$ SD, distributions, or boxplots (median, interquartile range, and 1.5 $\times$ IQR whiskers), as indicated. No data were excluded, and all representative images reflect results consistently observed across all acquired FOVs.

Code availability

Code to control the fluidics system is available at <https://github.com/fish-quant/autofish>. Code used for the analysis is available at <https://github.com/fish-quant/autofish-analysis>.

Data availability

The numerical source data underlying all graphs and charts in this study are publicly available in the Zenodo repository at <https://doi.org/10.5281/zenodo.19047523>. These data are organized by figure and panel as individual CSV files. The raw image data supporting the findings of this study are available from the corresponding author upon reasonable request due to their large file size.

Competing interests

The authors declare no competing interests.

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Author Contributions

C.W. performed wet-lab experiments and contributed to the development of the fluidics system. T.D. and J.B. performed data analysis. M.L. provided microscopy expertise and support. C.S. contributed to the development of the fluidics system. H.L. and A.M. provided mouse samples. M.I.G. contributed to the writing of the manuscript. J.-A.L.-V. and C.F. provided technical insights and supervised the mouse-related work. T.W. contributed to data analysis. F.M. conceived the study, implemented the fluidics system, and led the writing and coordination of the manuscript. All authors reviewed and approved the final version of the manuscript.

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Figures legends

Figure 1. Summary of the autoFISH workflow, from wetlab to analysis. **A** Schematic of the experimental workflow. Used oligonucleotides: Overview of primary, readout, imager, and stripping oligos and their nomenclature. Sequential FISH: Primary oligos are hybridized once; across N rounds, readout oligos reveal defined RNA subsets, while stripping oligos remove signals from previous rounds. SABER: Signal amplification is achieved via DNA concatemers, providing multiple binding sites for fluorescent imagers. Clearing: RNA targets are anchored to a polymer, and the sample is cleared to reduce autofluorescence and nonspecific background. **B** Overview of the fluidics system with the different components. **C** Samples can be placed in an imaging chamber that integrates fluidic inputs with high-quality microscopy. **D** The Python-based control software coordinates reagent delivery from the fluidics system with the microscope's acquisition sequences. **E** Provided software enables the analysis of different hybridization rounds and yields RNA abundance measurements for each segmented cell. The figure was created with BioRender.com.

Figure 2. Robustness of autoFISH over 20 Sequential Hybridization Rounds. **A** Images of the same cell with either *XPO1* and *KIF1C* for the respective first and last hybridization rounds. Please note that the nuclear background observed in *KIF1C* is due to nonspecific binding of certain primary oligonucleotides. The same intensity scaling [min, max] was used for images of each gene: *XPO1* [374, 2216], *KIF1C* [2334, 5708]. **B** Zooms in (yellow rectangles in A) showing *XPO1* (round 1) and *KIF1C* (round 2). **C** RNA counts per cell across rounds for *XPO1* and *KIF1C*. **D** Example of image registration with images from round 1 (purple) and round 19 (green). **E** Percentage of colocalized RNAs with reference to the first respective hybridization round (Distance threshold for co-localization = 0.5 μm). **F-G-H** Mean Signal over noise ratio (SNR), background intensity, and spots intensity across rounds. Number of field of views = 25; number of cells = 453; number of RNAs: *KIF1C* = 305K, *XPO1* = 55K. In panels C, F, G, H lines represent the median; shaded areas indicate the interquartile range (IQR, 25th–75th percentiles).

Figure 3. Signal amplification with SABER across hybridization rounds. **A** SABER amplifies a specific sequence by serving as a binding region for imaging oligos. Hybridization of these oligos amplifies the signal. The oligo also contains a toehold region (red) that permits probe stripping. **B** 2% agarose gel electrophoresis shows readout oligos with either 1 or 3 repeats, or SABER amplification with two durations (30 min and 2 h). Gel was visualized using a Fisher Bioblock Scientific UV imaging system. An image of the entire gel was captured on thermal paper and subsequently digitized and is shown with minimal cropping. **C** Hybridization conditions for the eight rounds (R1-R8): small, filled dots indicate a standard smFISH; larger dots indicate SABER FISH; x indicates stripping of the previous round. **D** Images of the same cell across different hybridization rounds. Images for the same gene are shown with the same intensity scaling. **E** RNA levels per cell across rounds. SABER runs are indicated with an asterisk. **F** Quantification of intensities of RNA and background across rounds. **G** Gene colocalization across imaging rounds (Distance threshold for co-localization = 0.2 μm). Colocalization of *XPO1* and *NFKBIA*. Spots were compared to reference detections from round 1 (*XPO1*) and round 2 (*NFKBIA*). Red:

same-gene colocalization; Black: cross-gene colocalization. Small points indicate colocalization coefficients for each field of view; bars show the pooled results across all fields. Number of fields of view = 25; number of cells = 58; number of RNAs: *NFKBIA* = 2.6K, *XPO1* = 6.5K. Boxplots: central line = median; box = Interquartile range (Q1–Q3); whiskers = max/min data points within 1.5× IQR. Outliers excluded.

Figure 4. Clearing reduces background and allows RNA detection in tissue. **A** Principle of clearing. RNAs can be anchored with epoxide and/or Poly-T oligos carrying an acrydite moiety. The panel was created with BioRender.com. **B** Influence of epoxide on nuclear signals. DAPI and smFISH images after clearing with Poly-T anchoring and epoxide. **C** Nucleus segmentation and RNA spots detection across rounds. **D** Results of analysis workflow from RNA detection to cell segmentation and cell type classification. The analysis was performed across 6 3D fields of view, resulting in the segmentation of 1.1K nuclei. RNA counts per gene: *Rtn2* (6.5K), *Lamp3* (1.0K), *Pecam1* (7.4K), *Ptpb* (12.8K), *Pdgfra* (3.8K), and *Chil3* (2.6). The yellow rectangle shows a region provided with a zoom-in. **E** Louvain clustering of the spatially resolved RNA profiles. Cell-type mapping based on our reference scRNAseq data by matching the cosine distance between cluster centroids.

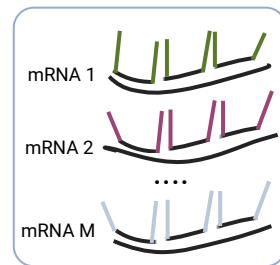
A Experimental approaches

Used oligonucleotides

Primary oligos	Readout oligos
	Complement imager Toehold Complement readout
Imager oligos	Stripping oligos

Sequential smFISH

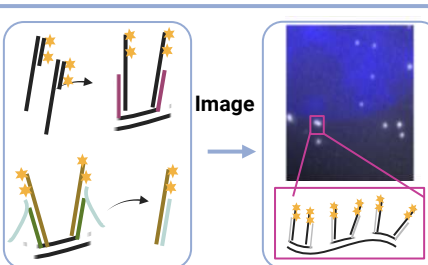
Primary hybridization: 12-36h



Hybridize Round N

Strip Round N-1

Readout hybridization/stripping: 1-3 h

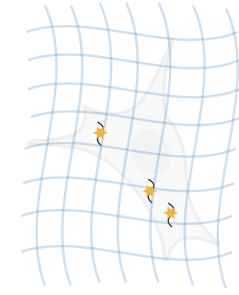


Image

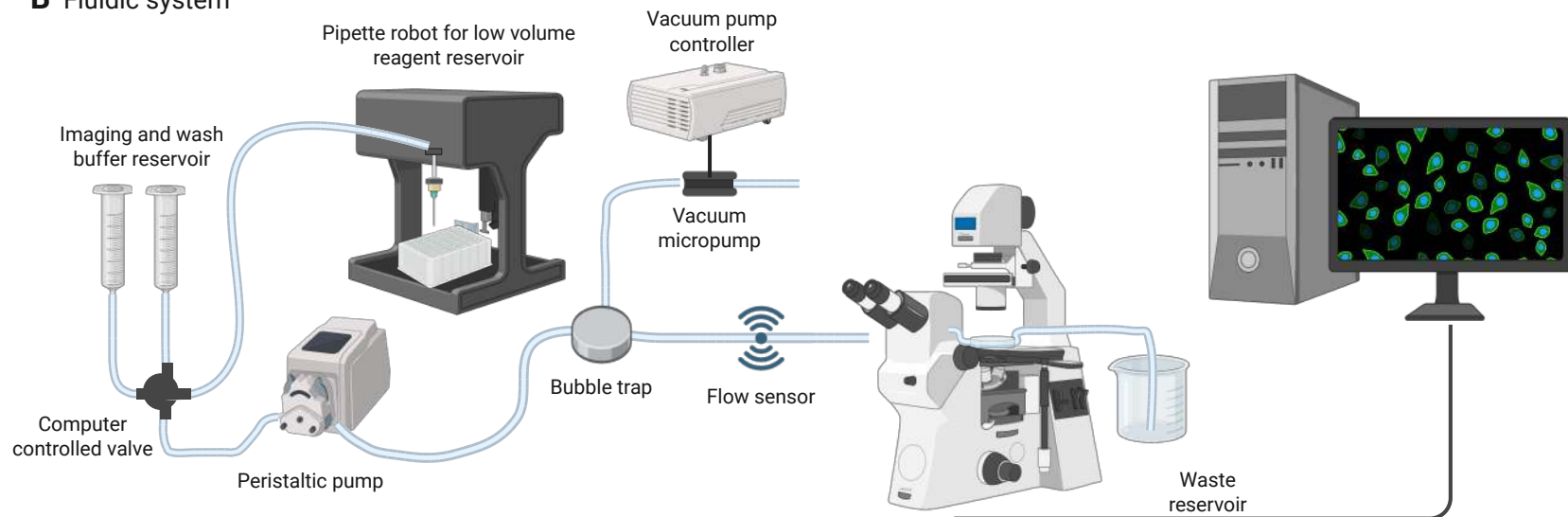
SABER



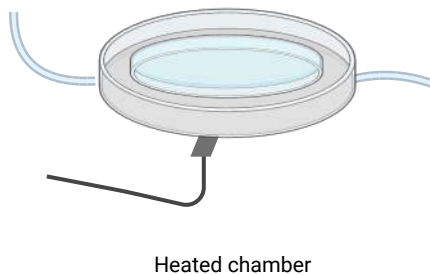
Clearing



B Fluidic system



C Sample setup



D Instrumentation control



E Data analysis

